

Fleischner's theorem has a natural extension to infinite graphs, which is much harder to prove:

Theorem 10.3.3. (Georgakopoulos 2009)

The square of every 2-connected locally finite graph contains a Hamilton circle.

We close the chapter with a far-reaching conjecture generalizing Dirac's theorem:

Conjecture. (Seymour 1974)

Let G be a graph of order $n \geq 3$, and let k be a positive integer. If G has minimum degree

$$\delta(G) \geq \frac{k}{k+1}n,$$

then G has a Hamilton cycle H such that $H^k \subseteq G$.

For $k = 1$, this is precisely Dirac's theorem. The conjecture was proved for large enough n (depending on k) by Komlós, Sárközy and Szemerédi (1998).

Exercises

1. An oriented complete graph is called a *tournament*. Show that every tournament contains a (directed) Hamilton path.
2. Show that every uniquely 3-edge-colourable cubic graph is hamiltonian. ('Unique' means that all 3-edge-colourings induce the same edge partition.)
3. Given an even positive integer k , construct for every $n \geq k$ a k -regular graph of order $2n + 1$.
4. Prove or disprove the following strengthening of Proposition 10.1.2: 'Every k -connected graph G with $|G| \geq 3$ and $\chi(G) \geq |G|/k$ has a Hamilton cycle.'
5. Let G be a graph, and $H := L(G)$ its line graph.
 - (i) Show that H is hamiltonian if G has a spanning Eulerian subgraph.
 - (ii)⁺ Deduce that H is hamiltonian if G is 4-edge-connected.
6. (i)⁻ Show that hamiltonian graphs are 1-tough.
 (ii) Find a graph that is 1-tough but not hamiltonian.
7. Prove the toughness conjecture for planar graphs. Does it hold with $t = 2$, or even with some $t < 2$?

- 8.⁻ Find a hamiltonian graph whose degree sequence is not hamiltonian.
- 9.⁻ Let G be a graph with fewer than i vertices of degree at most i , for every $i < |G|/2$. Use Chvátal's theorem to show that G is hamiltonian. (Thus in particular, Chvátal's theorem implies Dirac's theorem.)
10. Prove that the square G^2 of a k -connected graph G is k -tough. Use this to deduce Fleischner's theorem for graphs satisfying the toughness conjecture with $t = 2$.
11. Show that Exercise 6(i) has the following weak converse: for every non-hamiltonian graph G there exists a graph G' that has a pointwise greater degree-sequence than G but is not 1-tough.
12. (i) Show that, unlike the graphs satisfying Dirac's condition of $\delta \geq n/2$, graphs satisfying the degree condition of Theorem 10.1.3 can be sparse: there exists an integer d for which there are arbitrarily large graphs of average degree at most d that satisfy the condition.
(ii) Show that there is no integer d that bounds the average degrees of arbitrarily large graphs satisfying Chvátal's degree condition.
13. Find a connected graph G whose square G^2 has no Hamilton cycle.
- 14.⁻ Deduce from the proof of Fleischner's theorem that the square of a 2-connected graph contains a Hamilton path between any two vertices.
15. Show by induction on $|G|$ that the third power G^3 of any connected graph G of order at least 3 contains a Hamilton cycle.
- 16.⁺ Let G be a graph in which every vertex has odd degree. Show that every edge of G lies on an even number of Hamilton cycles.
(Hint. Let $xy \in E(G)$ be given. The Hamilton cycles through xy correspond to the Hamilton paths in $G - xy$ from x to y . Consider the set $\mathcal{H}$ of all Hamilton paths in $G - xy$ starting at x , and show that an even number of these end in y . To show this, define a graph on $\mathcal{H}$ so that the desired assertion follows from Proposition 1.2.1.)

Notes

The problem of finding a Hamilton cycle in a graph has the same kind of origin as its Euler tour counterpart and the four colour problem: all three problems come from mathematical puzzles older than graph theory itself. What began as a game invented by W.R. Hamilton in 1857 – in which ‘Hamilton cycles’ had to be found on the graph of the dodecahedron – re-emerged over a hundred years later as a combinatorial optimization problem of prime importance: the *travelling salesman problem*. Here, a salesman has to visit a number of customers, and his problem is to arrange these in a suitable circular route. (For reasons not included in the mathematical brief, the route has to be such that after visiting a customer the salesman does not pass through that town again.) Much of the motivation for considering Hamilton cycles comes from variations of this algorithmic problem.